Lab 3: Introduction to C Programming

Completed Report

Course:	SE1012 – Programming Methodology (Year 1, Semester 1)
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Overview

This document contains the solutions and explanations for Lab 3, focusing on C programming basics: data types, arithmetic expressions, console input/output (`printf` and `scanf`), and mathematical problem-solving on Linux machines.

1 Exercise 1: Design of a Hydroelectric Dam

Objective

Calculate the electrical power generated, in megawatts (MW), by a hydroelectric dam given its height and water flow rate, assuming 90% efficiency ($\eta = 0.9$).

Source Code (`exercise1.c`)

```c
#include <stdio.h>

int main() {
    double height, flow, mass, work, power;
    const double EFFICIENCY = 0.9;
    const double GRAVITY = 9.80;

    // Prompt user for input
    printf("Enter dam height (m): ");
    scanf("%lf", &height);
    printf("Enter water flow rate (m^3/s): ");
    scanf("%lf", &flow);

    // Calculations
    mass = flow * 1000.0;    // kg per second (1 m^3 = 1000 kg)
    work = mass * GRAVITY * height;    // m*g*h
    power = (EFFICIENCY * work) / 1000000.0;    // Watts to Megawatts

    printf("Predicted power generated: %.2f megawatts\n", power);
    return 0;
}
```

Explanation & Verification

- **Mass calculation:** One cubic metre of water weighs 1000 kg, so multiplying the flow rate by 1000 gives the mass flowing per second.

- **Work and power:** The work done per second is the gravitational potential energy mgh. Power is obtained by applying the 90% efficiency factor and dividing by 10^6 to convert watts to megawatts:

$$P = \frac{0.9 \times (\rho Q) \times g \times h}{10^6} \text{ MW}.$$

- **Test case verification:** For $h = 170\,\text{m}$ and $Q = 1.30 \times 10^3\,\text{m}^3/\text{s}$:

$$P = \frac{0.9 \times 1.30 \times 10^6 \times 9.80 \times 170}{10^6} = \mathbf{1949.22}\ \text{MW}.$$

A second run with $h = 100\,\text{m}$ and $Q = 800\,\text{m}^3/\text{s}$ gives 705.60 MW, matching the program output below.

```
readme
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/EX2$ vim exercise2.c
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/EX2$ cd ex1
-bash: cd: ex1: No such file or directory
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/EX2$ cd ..
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578$ cd ex1
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/ex1$ ls
execercise1  exercise1.c  readme
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/ex1$ ./execercise1
Enter dam height (m): 170
Enter water flow rate (m^3/s): 1300
Predicted power generated: 1949.22 megawatts
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/ex1$ ./execercise1
Enter dam height (m): 100
Enter water flow rate (m^3/s): 800
Predicted power generated: 705.60 megawatts
```

Figure 1: Program output for Exercise 1.

2 Exercise 2: A Jet Fighter Launched from an Aircraft Carrier

Objective

Calculate the constant acceleration (m/s^2) and the time (s) required for a jet fighter to reach its takeoff speed, given in km/hr, over a specified catapult distance in metres.

Source Code (`exercise2.c`)

```c
#include <stdio.h>

int main() {
    double speed_kmh, distance, v, a, t;

    // Prompt user for inputs with units
    printf("Enter takeoff speed (km/hr): ");
    scanf("%lf", &speed_kmh);
    printf("Enter catapult distance (meters): ");
    scanf("%lf", &distance);

    // Convert speed from km/hr to m/s
    v = speed_kmh / 3.6;

    // v = a*t and s = (1/2)*a*t^2  =>  a = v^2 / (2*s)
    a = (v * v) / (2.0 * distance);

    // Time t = v / a
    t = v / a;

    printf("Acceleration: %.2f m/s^2\n", a);
```

```c
    printf("Time to takeoff: %.2f seconds\n", t);
    return 0;
}
```

Explanation & Verification

- **Unit conversion:** The speed in km/hr is converted to m/s by dividing by 3.6.
- **Physics derivation:** Combining the equations of motion $v = at$ and $s = \frac{1}{2}at^2$ gives

$$a = \frac{v^2}{2s}, \qquad t = \frac{v}{a}.$$

- **Test case verification:** For 278 km/hr and 94 m: $v = 77.22\,\text{m/s}$, so $a = 31.72\,\text{m/s}^2$ and $t = 2.43\,\text{s}$, matching the output below.

```
compilation terminated.
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/ex2$ gcc exercise2.c -o exercise2
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/ex2$ ./exeecise2
-bash: ./exeecise2: No such file or directory
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/ex2$ ./ exercise2
-bash: ./: Is a directory
agais@ASHAN-PC:/mnt/c/Users/agais/OneDrive/Desktop/IT23632578/lab_3_IT23632578/ex2$ ./exercise2
Enter takeoff speed (km/hr): 278
Enter catapult distance (meters): 94
Acceleration: 31.72 m/s^2
Time to takeoff: 2.43 seconds
```

Figure 2: Compilation and program output for Exercise 2.

3 Exercise 3: Calculating the Base and Height of a Sail

Objective

Determine the base and height of a triangular boat sail from its area, given that the base is $\frac{2}{3}$ of the height.

Source Code (`exercise3.c`)

```c
#include <stdio.h>
#include <math.h>

int main() {
    double area, height, base;

    // Prompt user for sail area
    printf("Enter area of the sail (square meters): ");
    scanf("%lf", &area);

    // Area = 0.5 * base * height, and base = (2/3) * height
    // => Area = (1/3) * height^2  =>  height = sqrt(3 * Area)
    height = sqrt(area * 3.0);
    base = (2.0 / 3.0) * height;

    printf("Height of the sail: %.2f meters\n", height);
    printf("Base of the sail: %.2f meters\n", base);
    return 0;
}
```

Compile with the math library: `gcc exercise3.c -o exercise3 -lm`.

Explanation & Verification

- **Mathematical substitution:** Substituting base $= \frac{2}{3}$ height into Area $= \frac{1}{2} \times$ base $\times$ height gives Area $= \frac{1}{3}$ height^2.
- **Solving:** Rearranging yields height $= \sqrt{3 \times \text{Area}}$, computed with `sqrt()` from `<math.h>`.
- **Test case verification:** For an area of 75, height $= 15.00\,\text{m}$ and base $= 10.00\,\text{m}$. For an area of 108, height $= 18.00\,\text{m}$ and base $= 12.00\,\text{m}$.

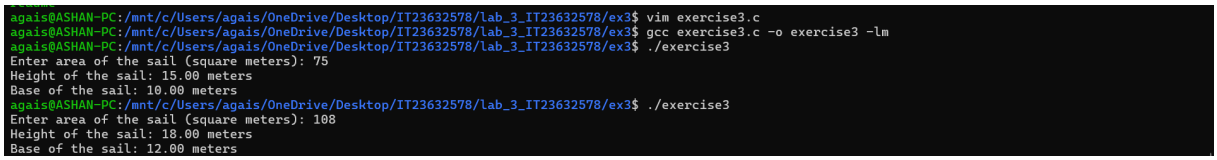


Figure 3: Program output for Exercise 3.